

Linear Algebra and Calculus Lecture 5

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Fall 2025

Systems of linear equations

A **linear equation** in the variables $x_1, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where b and the **coefficients** $a_1, \dots, a_n$ are real numbers (or complex numbers).

Remark. Equations like $x_1x_2 + 6x_2^3 = 0$ or $\sin x_3 - e^{x_1} = 2$ are NOT linear.

A **system of linear equations** (or a **linear system**) in the variables $x_1, \dots, x_n$ is a collection of linear equations in the variables $x_1, \dots, x_n$. For example we can consider the system of linear equations in x, y given by

$$\begin{aligned} 3x + 4y &= 5 \\ 2x - y &= 1. \end{aligned} \tag{1}$$

Both of these equations define a line in $\mathbb{R}^2$ (the plane).

Exercise. You are doing a software project consisting of three tasks: UI design, writing back-end code, and conducting tests. The project time has the following constraints:

- The project must be completed in exactly 120 hours.
- The time spent on UI design is twice the time spent on testing.
- The time spent on writing the back-end code is 20 hours more than the time spent on testing.

Let x_1 represent the time (in hours) spent designing the UI, let x_2 represent the time (in hours) spent writing the back-end code, and let x_3 represent the time (in hours) spent conducting tests. *Write a system of linear equations in the variables x_1, x_2, x_3 expressing the constraints listed above.*

Solutions

A **solution** of a system of linear equations in the variables $x_1, \dots, x_n$ is a list $(s_1, \dots, s_n)$ of real numbers (or complex numbers) for which each equation in the system holds true after substituting $s_1, \dots, s_n$ for $x_1, \dots, x_n$. The **solution set** is the set of all solutions.

Question 1. *What is the solution set in the example in equation (1)? Does every system of linear equations have a solution?*

If we picture a system of two equations in two variables as two lines in the plane, and a solution as their intersection, then we can conclude that a linear system has

- No solution (and we call it **inconsistent**), or
- exactly one solution (**consistent**), or
- infinitely many solutions (**consistent**).

Matrix form

We commonly write systems of linear equations in matrix form, for example the system

$$\begin{array}{rcl} x_1 - 2x_2 + x_3 & = & 0 \\ 2x_2 - 8x_3 & = & 8 \\ 5x_1 & - & 5x_3 = 10 \end{array}$$

has **coefficient matrix** (of size 3×3)

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix},$$

and **augmented matrix** (of size 3×4)

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}.$$

A matrix of size $m \times n$ has m rows and n columns.

Exercise. Find a solution for the previous system.

We commonly solve the equations by applying elementary row operations to the matrix, namely

1. Replacement of one row by the sum of itself and a multiple of another row.
2. Interchange of two rows.
3. Scaling of all entries in a row by a non-zero constant.

Proposition. *If the augmented matrices of two linear systems are row equivalent (we can perform row operations to obtain the other), then the two systems have the same solution set.*

Echelon matrix

Definition. We say that a matrix is in **echelon form** if

1. All non-zero rows are above any rows of all zeroes.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

and that it is in **reduced echelon form** if moreover

4. The leading entry in each nonzero row is 1.
5. Each leading 1 is the only nonzero entry in its column.

Exercise. Turn the system of linear equations from Exercise 1 into a matrix and apply elementary row operators to put the matrix into reduced row echelon form. What are the solutions to our original problem?

Theorem. (*Uniqueness of the Reduced Echelon Form*) *Each matrix is row equivalent to one and only one reduced echelon matrix.*

Definition. A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A . A **pivot column** is a column of A that contains a pivot position.

Assume we have the following matrix already in reduced echelon form

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The associated system is

$$\begin{array}{rcl} x_1 & -5x_3 & = 1 \\ x_2 & +x_3 & = 4 \\ & & 0 = 0 \end{array}$$

The variables x_1 and x_2 corresponding to pivot columns in the matrix are called **basic variables**, and x_3 is called a **free variable**. We can write the solution as

$$\begin{cases} x_1 = 1 + 5x_3 \\ x_2 = 4 - x_3 \\ x_3 \text{ is free} \end{cases}$$

Example. Consider a system whose augmented matrix has been reduced to

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

which reduces further to

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

with associated system

$$\begin{aligned} x_1 + 6x_2 + 3x_4 &= 0 \\ x_3 - 4x_4 &= 5 \\ x_5 &= 7 \end{aligned}$$

which has general solution

$$\begin{cases} x_1 = -6x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases}$$

But what if instead our matrix reduced to

$$\begin{bmatrix} 1 & 6 & 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 7 \end{bmatrix}$$

There is no solution!

Theorem. (*Existence and Uniqueness Theorem*) A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column—that is, if and only if an echelon form of the augmented matrix has no row of the form

$$[0 \quad \cdots \quad 0 \quad b] \quad \text{with } b \text{ non-zero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.